



Badri Vishal Kasuba

MS by Research Scholar

Indian Institute of Technology Bombay

Mobile No: +91 89853 91074

Email ID: kasubabadri@gmail.com

GitHub ID: kasuba-badri-vishal

Program	Specialization	Institute	Year	GPA/%
Masters by Research	Computer Science and Engineering	IIT Bombay	2022-25	9.03/10
Bachelors in Engineering	Computer Science and Engineering	BITS-Pilani	2017-21	7.91/10

WORK EXPERIENCE

- **Quadric IO | Senior Compiler Engineer** (Jul'25 - Present)
 - Deep Learning compiler design and development of Quadric Graph Compiler (CGC) through TVM optimizations
 - Worked on addressing and maintaining the Quadric Chimera Chip CGC issues and feature addition in TVM
- **Amazon | Software Development Engineer** (Nov'21 - Jul'22)
 - Involved in Digital Payments team, worked on **accelerating payment experience** for the Amazon partners
 - Developed and deployed **4 new payment processor plugins** for internal clients impacting payment services.
 - Maintained and updated codebase with new features and CRUD operations for payment plugins
- **ZS Associates | Data Science Associate** (May'21 - Nov'21)
 - Worked on real-life pharma-based projects on **data analytics** for providing solutions to ZS clients
 - Implemented AI models for data analytics and provided effective predictions required for product delivery
 - Worked on weak-supervision techniques and NLP tasks like NER, Parsing, etc in the realm of medical data

PUBLICATIONS

- **DRISHTIKON**: Multi-Granular Visual Grounding for Text-Rich Documents (preprint, 2025)
- **PLATTER**: A Page-level Handwritten Text Recognition System for Indic Scripts (Journal preprint, 2025)
- **SPRINT**: Script-agnostic Structure Recognition in Tables (ICDAR, 2024)
- **TEXTRON**: Weakly Supervised Multilingual Text Detection through Data Programming (WACV, 2024)

RESEARCH PROJECTS

- **Computer Vision Research Scholar | BharatGen** (Nov'24 - Jun'25)
(Research Thesis | Guide: Prof. Ganesh Ramakrishnan)
 - Contributing to Foundational Multimodal model development for Document Understanding in Indic documents.
 - Leading the Vision team for Indic Circulars VQA, focusing on robust model development and Q&A data synthesis.
- **Handwritten Text Recognition Research in Indian Languages** (Jan'24 - Present)
(Research Thesis | Guide: Prof. Ganesh Ramakrishnan and Prof. Parag Chaudhuri)
 - Developed robust Text Recognition models across **10** Indian languages with multilingual support
 - Conducted in-depth comparative analysis for research, delivering valuable insights and optimizing performance
 - Seamlessly integrated developed models into the BHASHINI API platform, facilitating improved accessibility.

SCHOLASTIC ACHIEVEMENTS & EXTRA-CURRICULAR ACTIVITIES

- Achieved GATE CSE 2022: **99.36** percentile | GATE CSE 2021: **98.82** percentile | BITSAT 2017: **340/450** marks
- Awarded with **Certificate of Excellence in Research** for Masters programme, Department of CSE, IIT Bombay (2025)
- Awarded with **Best Poster Presentation** at **RISC 2024**, Research Symposium, Department of CSE, IIT Bombay (2024)
- Led the IIT Bombay team in digitizing books for **CSTT**, Ministry of Education Department, Govt. of India (2023)
- Participated and Presented research findings at **NCVPRIPG 2023**, IIT Jodhpur as part of NLTM-Bhashini (2023)
- Won the **Best Sports Secretary Award** for PG Sports 2023-24 at IIT Bombay for enhancing sporting culture (2024)
- Organized one-day Workshop on Improvizng Document Analysis tools at **ICFHR 2022**, IIIT-Hyderabad (2022)
- Participated in the Summer Projects and Workshop on **Android App Development** held at IIT Madras (2018)

KEY ACADEMIC PROJECTS AND ASSIGNMENTS

- **Effect of Enforcing Oversmoothing in Transformers** (Spring 2024)
(CS 769: Optimization in Machine Learning | Instructor: **Prof. Ganesh Ramakrishnan**)
 - Investigated Oversmoothing phenomenon in transformer architectures.
 - Contributed to a deeper understanding of language understanding through transformer analysis.
- **Digitization of Ancient Indian Manuscripts** (Autumn 2023)
(CS 663: Digital Image Processing | Instructor: **Prof. Ajit Rajwade**)
 - Developed image processing techniques to enhance representation of complex manuscript document images
 - Established a foundation for future research in manuscript digitization and preservation
- **DocToSpeech Application** (Spring 2023)
(CS 772: Deep Learning for Natural Language Processing | Instructor: **Prof. Pushpak Bhattacharya**)
 - Developed a comprehensive pipeline integrating OCR, summarization, machine translation, and TTS components
 - Optimized pipeline performance for real-world applications and downstream tasks for individual use.
- **Action Adventure Game with Python** (Autumn 2022)
(CS 699: Software Lab | Instructor: **Prof. Bhaskaran Raman**)
 - Developed an engaging Action Adventure Game, featuring a space ranger character and interactive gameplay
 - Wrote over 400 lines of Python code, using object-oriented programming principles for game entities
 - Achieved project Excellence remark with 22/20 marks for the work done
- **Voxelized and FMX Modeling and Animation** (Autumn 2022)
(CS 675: Computer Graphics | Instructor: **Prof. Parag Chaudhuri**)
 - Developed a high-accuracy lightweight crop disease detection model surpassing InceptionV3 and Xception by 6%.
 - Leveraged potato and tomato dataset to provide solution to aid effective crop disease identification for farmers.
- **Tag Prediction for Stack Overflow Questions** (Autumn 2022)
(CS 725: Foundations of Machine Learning | Instructor: **Prof. Preethi Jyothi**)
 - Developed a high-accuracy lightweight crop disease detection model surpassing InceptionV3 and Xception by 6%.
 - Leveraged potato and tomato dataset to provide solution to aid effective crop disease identification for farmers.

TECHNICAL SKILLS

- **Programming & Scripting Languages:** C, C++, Java, Python, Bash Scripting, JavaScript, SQL
- **Tools & Libraries:** Huggingface, Pandas, Numpy, Matplotlib, OpenCV, PyTorch, Postman, Git, Docker, Flask, ReactJS

COURSES TAKEN

- **DataScience:** Foundations of Machine Learning, Optimization in Machine Learning, Deep Learning with NLP, Computer Vision, Automated Speech Recognition, Software Engineering
- **Graphics:** Digital Image Processing, Computer Graphics

POSITIONS OF RESPONSIBILITY

- **Research Assistant through Project | National Language Translation Mission - BHASHINI Project** 2022-Present
 - Developed and Delivered 13 printed and 10 handwritten text recognition model APIs to Bhashini for public use
 - Mentored and Guided more than 15 research interns at IITB LEAP-OCR team for developing ocr applications
 - Contributed APIs to OCR applications for Indian languages to Bhashini, an initiative of **MeitY, Govt. of India**
- **PG Sports Secretary | Computer Science Department** (2023-24)
 - Served as department sports secretary and improved the sporting culture of the CSE Department
 - Led the department to achieve **Runners-up** title in the overall PG General Championship 2023-24
 - Recognized for enhancing the department's sporting culture and spearheading innovative initiatives
- **Student Companion | Institute Student Companion Program** (2023-24)
 - Mentored new entrants, facilitating smooth onboarding and guidance for academic success at the Institute

HOBBIES & SKILLS

- **Sports:** Swimming, Cricket, Chess and Table Tennis
- **Languages:** Proficient in Telugu, Hindi, English